

Course Name	General objective/Competency	Specific objectives /learning outcomes	Learning Outcome						
			Cognitive						
			Remembering	Understanding	Application	Analysis	Evaluation	Creation/Synthesis	No. of items
OOP in Java	Develop the necessary skills in designing, coding, debugging, and documenting large programs using Java programming languages	Students will be able to explain the basic object-oriented concepts		3					3
		Students will be able to identify the differences between structural and object-oriented programming paradigms				2			2
		Students will be able to use java programming language to code, debug and execute results			3				3

The statement "**Students will be able to explain the basic object-oriented concepts**" is a learning outcome that falls under the Comprehension level of the cognitive domain. **Comprehension (understanding)** refers to the ability to understand the meaning of information, and in this case, students are expected to understand the basic concepts of object-oriented programming.

To achieve this learning outcome, students should be able to understand and explain the following basic object-oriented concepts:

1. **Classes and Objects:** Students should understand the difference between classes and objects. A class is a blueprint for creating objects, while an object is an instance of a class.

Example: A student should be able to explain that a class is like a blueprint for creating objects, and an object is like a house built from the blueprint.

2. **Encapsulation:** Students should understand how encapsulation is used to group related data and functions into a single unit called a class, and how this helps to organize code and make it more modular.

Example: A student should be able to explain that encapsulation is like putting related items in a box, which makes it easier to organize and find the items.

3. **Inheritance:** Students should understand how inheritance allows a new class to be based on an existing class, inheriting properties and methods from the existing class.

Example: A student should be able to explain that inheritance is like a child inheriting traits from their parents, such as eye color or height.

4. **Polymorphism:** Students should understand how polymorphism allows objects of different classes to be treated as if they are of the same class, and how this helps to make code more flexible and adaptable.

Example: A student should be able to explain that polymorphism is like a person who can speak multiple languages and can communicate with people from different countries.

5. **Abstraction:** Students should understand how abstraction involves hiding unnecessary details and presenting only the essential features of an object, and how this helps to simplify code and make it easier to understand.

Example: A student should be able to explain that abstraction is like a car dashboard that shows only the essential information, such as speed and fuel level, and hides other details like the engine's internal workings.

Overall, achieving comprehension of basic object-oriented concepts requires a deep understanding of each concept and how they relate to each other. Students should be able to explain each concept in their own words, provide examples of how each concept is used in real-world programming scenarios, and demonstrate an understanding of how each concept relates to other object-oriented concepts.

Sample multiple-choice questions based on basic object-oriented concepts.

1. What is a class in object-oriented programming?
 - A. An instance of an object
 - B. A blueprint for creating objects**
 - C. A function that performs a specific task
 - D. A variable that stores a value

2. What is an object in object-oriented programming?
 - A. A Blueprint for creating classes
 - B. An instance of a class**
 - C. A function that performs a specific task
 - D. A variable that stores a value
3. Which of the following is not a basic object-oriented concept?
 - A. Abstraction
 - B. Encapsulation
 - C. Functions**
 - D. Inheritance
4. What is encapsulation in object-oriented programming?
 - A. The process of hiding data and methods within a class**
 - B. The process of exposing data and methods to other classes
 - C. The process of breaking down a class into smaller parts
 - D. The process of combining multiple classes into a single class
5. What is inheritance in object-oriented programming?
 - A. The process of creating a new class based on an existing class**
 - B. The process of breaking down a class into smaller parts
 - C. The process of hiding data and methods within a class
 - D. The process of exposing data and methods to other classes
6. What is polymorphism in object-oriented programming?
 - A. The process of hiding data and methods within a class
 - B. The process of combining multiple classes into a single class
 - C. The process of treating objects of different classes as if they are of the same class**
 - D. The process of breaking down a class into smaller parts
7. What is an abstraction in object-oriented programming?
 - A. The process of exposing data and methods to other classes
 - B. The process of breaking down a class into smaller parts
 - C. The process of hiding unnecessary details and presenting only the essential features of an object**
 - D. The process of combining multiple classes into a single class

8. Which of the following is an example of encapsulation?
- A. A class that inherits properties and methods from another class
 - B. A class that exposes all of its variables and methods to other classes
 - C. A class that groups related variables and function into a single unit**
 - D. A class that treats objects of different classes as if they are of the same class
9. Which of the following is an example of inheritance?
- A. A class that groups related variables and function into a single unit
 - B. A class that exposes all of its variables and methods to other classes
 - C. A class that inherits properties and methods from another class**
 - D. A class that treats objects of different classes as if they are of the same class
10. Which of the following is an example of polymorphism?
- A. A class that groups related variables and function into a single unit
 - B. A class that inherits properties and methods from another class
 - C. A class that treats objects of different classes as if they are of the same class**
 - D. A class that hides unnecessary details and presents only the essential features of an object
11. Which of the following is an example of abstraction?
- A. A class that groups related variables and function into a single unit
 - B. A class that exposes all of its variables and methods to other classes
 - C. A class that hides unnecessary details and presents only the essential features of an object**
 - D. A class that treats objects of different classes as if they are of the same class
12. What is the main benefit of encapsulation in object-oriented programming?
- A. It allows a new class to be based on an existing class
 - B. It allows objects of different classes to be treated as if they are of the same class
 - C. It helps to organize code and make it more modular**
 - D. It helps to make code more flexible and adaptable
13. What is the main benefit of inheritance in object-oriented programming?
- A. It helps to organize code and make it more modular
 - B. It helps to make code more flexible and adaptable
 - C. It reduces code duplication and makes it easier to maintain code**
 - D. It allows objects of different classes to be treated as if they are of the same class

14. What is the main benefit of polymorphism in object-oriented programming?
- A. It helps to organize code and make it more modular
 - B. It allows a new class to be based on an existing class
 - C. It allows objects of different classes to be treated as if they are of the same class
 - D. It makes code more flexible and adaptable**
15. What is the main benefit of abstraction in object-oriented programming?
- A. It reduces code duplication and makes it easier to maintain code
 - B. It helps to make code more flexible and adaptable
 - C. It simplifies code and makes it easier to understand**
 - D. It allows objects of different classes to be treated as if they are of the same class
16. Which of the following is not an example of abstraction?
- A. A car dashboard that shows only the essential information
 - B. A class that groups related variables and function into a single unit
 - C. A class that exposes all of its variables and methods to other classes
 - D. A class that treats objects of different classes as if they are of the same class**
17. Which of the following is not an example of polymorphism?
- A. A person who can speak multiple languages and can communicate with people from different countries
 - B. A class that can be based on an existing class
 - C. A function that can perform multiple tasks depending on the context**
 - D. An object that can be treated as if it is of a different class
18. Which of the following is not an example of inheritance?
- A. A subclass of a vehicle class that inherits properties and methods related to all vehicles**
 - B. A child class that inherits properties and methods from a parent class
 - C. A new class that is created based on an existing class
 - D. A class that treats objects of different classes as if they are of the same class
19. What is the difference between a class and an object in object-oriented programming?
- A. A class is a blueprint for creating objects, while an object is an instance of a class**
 - B. A class is an instance of an object, while an object is a blueprint for creating classes
 - C. A class and an object are the same things

D. A class is a function that performs a specific task, while an object is a variable that stores a value

20. What is the main purpose of object-oriented programming?

- A. To organize code and make it more modular
- B. To reduce code duplication and make it easier to maintain code
- C. To make code more flexible and adaptable
- D. All of the above**

The statement "**Students will be able to identify the differences between structural and object-oriented programming paradigms**" is a learning outcome that falls under the Analysis level of the cognitive domain. Analysis involves breaking down information into parts and examining the relationships between them. In this case, students are expected to analyze the differences between two programming paradigms.

To achieve this learning outcome, students should be able to identify and analyze the following differences between structural and object-oriented programming paradigms:

- 1. Data and Functionality:** Structural programming focuses on dividing a program into smaller functions and processing data using these functions. Object-oriented programming, on the other hand, focuses on encapsulating data and functions into objects.

Example: A student should be able to explain that in structural programming, a program can be divided into functions that perform a specific task, such as sorting or searching data. In object-oriented programming, data and functionality are encapsulated into objects, which can be used to represent real-world objects like cars, people, or buildings.

- 2. Code Reusability:** Object-oriented programming allows code to be reused through inheritance and polymorphism, while structural programming relies on functions and procedures to perform specific tasks.

Example: A student should be able to explain that in object-oriented programming, code can be reused through inheritance, which allows a class to inherit properties and methods from another class. Polymorphism also allows objects of different classes to be treated as if they are of the same class. In contrast, structural programming relies on

functions and procedures to perform specific tasks, with little to no code reuse between them.

3. **Modularity:** Object-oriented programming is more modular than structural programming, due to its encapsulation and inheritance features, which allow for greater code organization and flexibility.

Example: A student should be able to explain that encapsulation in object-oriented programming allows related data and functions to be grouped into a single unit, making it easier to organize and manage code. Inheritance also allows for greater modularity, as it allows a new class to be based on an existing class, inheriting its properties and methods. In contrast, structural programming relies on functions and procedures that may not be as modular or organized.

4. **Focus on Objects:** Object-oriented programming is centred around objects, which represent real-world entities, while structural programming is focused on functions and procedures that operate on data.

Example: A student should be able to explain that object-oriented programming focuses on objects, which represent real-world entities and have properties and behaviours. These objects can be manipulated and processed using methods, which are functions that are part of the object. In contrast, structural programming focuses on functions and procedures that operate on data, with little emphasis on the objects themselves.

Overall, achieving analysis of the differences between structural and object-oriented programming paradigms requires a deep understanding of each paradigm and how they differ from each other. Students should be able to analyze and compare the features of each paradigm, provide examples of how each paradigm is used in real-world programming scenarios, and evaluate the strengths and weaknesses of each paradigm for different types of programming tasks.

For example, a student may analyze the benefits of using object-oriented programming for developing complex software systems, while also examining the limitations of structural programming for such tasks. Additionally, students should be able to apply their knowledge of each paradigm to design and implement code using appropriate programming techniques and best practices.

Sample multiple-choice questions on identifying the differences between structural and object-oriented programming paradigms

1. What is the main difference between structural and object-oriented programming paradigms?
 - A. Structural programming focuses on objects, while object-oriented programming focuses on functions.
 - B. Structural programming relies on functions and procedures, while object-oriented programming encapsulates data and functions into objects.**
 - C. Structural programming emphasizes code reusability, while object-oriented programming focuses on modularity.
 - D. Structural programming is more modular than object-oriented programming.
2. Which programming paradigm is more focused on objects?
 - A. Structural programming
 - B. Object-oriented programming (Answer)
 - C. Both programming paradigms are equally focused on objects.
 - D. Neither programming paradigm is focused on objects.
3. How does object-oriented programming allow code to be reused?
 - A. Through functions and procedures
 - B. Through polymorphism and inheritance**
 - C. Through encapsulation
 - D. Through modularity
4. What is encapsulation in object-oriented programming?
 - A. The process of dividing a program into smaller functions and processing data using these functions
 - B. The process of encapsulating data and functions into objects**
 - C. The process of reusing code through inheritance and polymorphism
 - D. The process of organizing code into modules
5. What is inheritance in object-oriented programming?
 - A. The process of dividing a program into smaller functions and processing data using these functions
 - B. The process of encapsulating data and functions into objects

- C. The process of reusing code through inheritance and polymorphism**
- D. The process of organizing code into modules
6. Which programming paradigm is more modular?
- A. Structural programming
- B. Object-oriented programming**
- C. Both programming paradigms are equally modular.
- D. Neither programming paradigm is modular.
7. What is the main benefit of encapsulation in object-oriented programming?
- A. It allows code to be reused through inheritance and polymorphism.
- B. It allows for greater modularity and organization of code.**
- C. It allows for greater code flexibility and adaptability.
- D. It allows for greater code efficiency.
8. What is the main benefit of inheritance in object-oriented programming?
- A. It allows code to be reused through encapsulation.
- B. It allows for greater modularity and organization of code.
- C. It allows objects of different classes to be treated as if they are of the same class.
- D. It allows code to be reused through inheritance and polymorphism.**
9. Which programming paradigm relies on functions and procedures to perform specific tasks?
- A. Structural programming**
- B. Object-oriented programming
- C. Both programming paradigms rely on functions and procedures to perform specific tasks.
- D. Neither programming paradigm relies on functions and procedures to perform specific tasks.
10. Which programming paradigm is more focused on dividing a program into smaller functions?
- A. Structural programming**
- B. Object-oriented programming
- C. Both programming paradigms are equally focused on dividing a program into smaller functions.

- D. Neither programming paradigm is focused on dividing a program into smaller functions.
11. Which programming paradigm is more focused on encapsulating data and functionality into objects?
- A. Structural programming
 - B. Object-oriented programming**
 - C. Both programming paradigms are equally focused on encapsulating data and functionality into objects.
 - D. Neither programming paradigm is focused on encapsulating data and functionality into objects.
12. Which programming paradigm is more focused on code reusability?
- A. Structural programming
 - B. Object-oriented programming**
 - C. Both programming paradigms are equally focused on code reusability.
 - D. Neither programming paradigm is focused on code reusability.
13. Which programming paradigm is more suited for developing complex software systems?
- A. Structural programming
 - B. Object-oriented programming**
 - C. Both programming paradigms are equally suited for developing complex software systems.
 - D. Neither programming paradigm is suited for developing complex software systems.
14. Which programming paradigm is more flexible and adaptable?
- A. Structural programming
 - B. Object-oriented programming**
 - C. Both programming paradigms are equally flexible and adaptable.
 - D. Neither programming paradigm is flexible and adaptable.
15. Which programming paradigm is more focused on functions and procedures that operate on data?
- A. Structural programming**
 - B. Object-oriented programming
 - C. Both programming paradigms are equally focused on functions and procedures that operate on data.

- D. Neither programming paradigm is focused on functions and procedures that operate on data.
16. Which programming paradigm is more focused on objects that represent real-world entities?
- A. Structural programming
 - B. Object-oriented programming**
 - C. Both programming paradigms are equally focused on objects that represent real-world entities.
 - D. Neither programming paradigm is focused on objects that represent real-world entities.
17. Which programming paradigm is more focused on dividing code into modules?
- A. Structural programming
 - B. Object-oriented programming**
 - C. Both programming paradigms are equally focused on dividing code into modules.
 - D. Neither programming paradigm is focused on dividing code into modules.
18. Which programming paradigm allows for objects of different classes to be treated as if they are of the same class?
- A. Structural programming
 - B. Object-oriented programming
 - C. Both programming paradigms allow for objects of different classes to be treated as if they are of the same class.
 - D. Neither programming paradigm allows for objects of different classes to be treated as if they are of the same class.**
19. Which programming paradigm relies on encapsulation and inheritance to achieve code reusability?
- A. Structural programming
 - B. Object-oriented programming**
 - C. Both programming paradigms rely on encapsulation and inheritance to achieve code reusability.
 - D. Neither programming paradigm relies on encapsulation and inheritance to achieve code reusability.
20. Which programming paradigm is more focused on organizing code into smaller functions and procedures?

A. Structural programming

B. Object-oriented programming

C. Both programming paradigms are equally focused on organizing code into smaller functions and procedures.

D. Neither programming paradigm is focused on organizing code into smaller functions and procedures.

- **Students will be able to use Java programming language to code, debug, and execute results**

The statement "**Students will be able to use Java programming language to code, debug, and execute results**" is a learning outcome that falls under the Application level of the cognitive domain. Application involves the ability to apply knowledge and skills to solve problems or complete tasks. In this case, students are expected to apply their knowledge of Java programming to write, debug, and execute code.

To achieve this learning outcome, students should be able to:

- Understand the syntax and structure of Java programming language:
- Students should be able to read and write Java code
- Understand the different data types, control structures, and functions used in Java programming, and apply this knowledge to write efficient and effective code.

Example: A student should be able to write a Java program that calculates the average of a list of numbers using loops and conditional statements.

- **Debug Java code:** Students should be able to identify and resolve errors in Java code, understand common programming errors, and use debugging tools to identify and fix issues in their code.
- **Example:** A student should be able to identify and fix errors in a Java program that is returning incorrect results, using debugging tools such as breakpoints and logging.
- **Execute Java programs:** Students should be able to compile and execute Java programs, understand the different tools used to compile and execute Java code, and troubleshoot issues related to program execution.

Example: A student should be able to compile and execute a Java program that reads a file and prints the contents to the console.

- **Apply object-oriented programming principles:** Students should be able to apply object-oriented programming principles such as encapsulation, inheritance, and polymorphism to write efficient and maintainable Java code.

Example: A student should be able to use inheritance to create a subclass that inherits properties and methods from a superclass, and override methods in the subclass to modify its behavior.

- **Use Java libraries and frameworks:** Students should be able to use Java libraries and frameworks to simplify coding tasks, increase productivity, and enhance the functionality of their Java programs.

Example: A student should be able to use the JDBC (Java Database Connectivity) library to connect to a database and retrieve data using Java code.

Overall, achieving the application of Java programming language requires a deep understanding of the syntax, structure, and principles of Java programming, as well as the ability to apply this knowledge to write efficient and effective code. Students should be able to identify and resolve errors in their code execute Java programs, apply object-oriented principles, and use libraries and frameworks to enhance the functionality and productivity of their Java programs. For example, a student may use the spring framework to simplify the development of a web application, or use the JUnit library to write unit tests for their Java code.

Sample Questions related to using Java programming language to code, debug, and execute results

1. What is the first step in writing a Java program?
 - A. Debugging the code
 - B. Compiling the code
 - C. Writing the code**
 - D. Executing the code
2. What is the process of identifying and resolving errors in Java code called?
 - A. Debugging**
 - B. Compiling
 - C. Executing
 - D. Writing
3. What is the process of turning Java code into machine-readable instructions called?
 - A. Debugging

B. Compiling

C. Executing

D. Writing

4. What is the process of running a Java program called?

A. Debugging

B. Compiling

C. Executing

D. Writing

5. What is the purpose of using loops in Java programming?

A. To encapsulate data and functions into objects

B. To control the flow of code execution

C. To create a new data type

D. To allow objects of different classes to be treated as if they are of the same class

6. What is the purpose of conditional statements in Java programming?

A. To encapsulate data and functions into objects

B. To control the flow of code execution based on a condition

C. To create a new data type

D. To allow objects of different classes to be treated as if they are of the same class

7. What is the purpose of using arrays in Java programming?

A. To encapsulate data and functions into objects

B. To hold a collection of values of the same data type

C. To create a new data type

D. To allow objects of different classes to be treated as if they are of the same class

8. What is the purpose of using functions in Java programming?

A. To encapsulate data and functions into objects

B. To hold a collection of values of the same data type

C. To control the flow of code execution based on a condition

D. To execute a specific task or set of tasks

9. What is the purpose of using variables in Java programming?

A. To execute a specific task or set of tasks

B. To hold a collection of values of the same data type

C. To encapsulate data and functions into objects

D. To store and manipulate data

10. What is the purpose of using comments in Java programming?

A. To execute a specific task or set of tasks

B. To hold a collection of values of the same data type

C. To encapsulate data and functions into objects

D. To add notes and explanations to code

11. What is the purpose of using try-catch blocks in Java programming?

A. To store and manipulate data

B. To handle and recover from exceptions

C. To execute a specific task or set of tasks

D. To add notes and explanations to code

12. Which of the following statements accurately describes the Java programming language?

A. Java is a compiled language

B. Java is an interpreted language

C. Java is a scripting language

D. Java is a mark-up language

13. Which of the following is the correct syntax for declaring a variable in Java?

A. `int myVariable = 5;`

B. `myVariable int = 5;`

C. `variable myVariable = 5;`

D. `int = myVariable(5);`

14. Which of the following is the correct syntax to create a new instance of an object in Java?

A. `object = new Object();`

B. `Object = new object();`

C. `Object = new Object;`

D. `object = new object;`

15. What is the purpose of the "public static void main (String [] args)" method in Java?

A. It is the starting point of a Java program.

B. It is used to declare variables.

C. It is used to print output to the console.

D. It is used to create new objects.

16. Which of the following is the correct syntax of a for loop in Java?

- A. for i = 0; i < 10; i++
- B. **for (int i = 0; i <10; i++)**
- C. for (i = 0; i < 10; i++)
- D. for (int i < 10; i++)

17. Which of the following is the correct syntax for a while loop in Java?

- A. **while (i < 10) { }**
- B. while i < 10 { }
- C. while (i < 10);
- D. while i < 10: { }

18. Which of the following is the correct syntax for a conditional statement in Java?

- A. if (x = 5)
- B. **if (x == 5)**
- C. if x = 5
- D. if x == 5

19. Which of the following is the correct syntax to declare and initialize an array in Java?

- A. **int[] myArray = new int[5];**
- B. myArray[] = {1, 2, 3, 4, 5};
- C. int myArray[5] = {1, 2, 3, 4, 5};
- D. myArray = new int[5] {1, 2, 3, 4, 5};

20. Which of the following is the correct way to catch an exception in Java?

- A. try { } catch Exception { }
- B. try { } catch (Exception e) { }
- C. catch (Exception e) { } try { }
- D. catch Exception { } try { }

21. Which of the following is the correct way to print output to the console in Java?

- A. print("Hello World");
- B. **System.out.print("Hello World");**
- C. console.log("Hello World");
- D. printf("Hello World");